Model for “GaAs”

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General Condition

- Basis type: `lg`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: $[A_1, A_2, E, T_1, T_2]$
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: SG No. 216 $T_{\rm d}^2$ $F\bar{4}3m$ [cubic]
- Associated point group: PG No. 216 $T_{\rm d}$ $\bar{4}3m$ [cubic]
- Unit cell:
  $a =$ 1.00000, $b =$ 1.00000, $c =$ 1.00000, $\alpha =$ 90.0, $\beta =$ 90.0, $\gamma =$ 90.0
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ [1.00000, 0.00000, 0.00000]
  $\boldsymbol{a}_2 =$ [0.00000, 1.00000, 0.00000]
  $\boldsymbol{a}_3 =$ [0.00000, 0.00000, 1.00000]

- Plus sets:
  $+[0,\,0,\,0]$, $+[0,\,\frac{1}{2},\,\frac{1}{2}]$, $+[\frac{1}{2},\,0,\,\frac{1}{2}]$, $+[\frac{1}{2},\,\frac{1}{2},\,0]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1\vert 0\}$	2	$\{2_{001}\vert 0\}$	3	$\{2_{010}\vert 0\}$	4	$\{2_{100}\vert 0\}$	5	$\{3^{+}_{111}\vert 0\}$
6	$\{3^{+}_{-11-1}\vert 0\}$	7	$\{3^{+}_{1-1-1}\vert 0\}$	8	$\{3^{+}_{-1-11}\vert 0\}$	9	$\{3^{-}_{111}\vert 0\}$	10	$\{3^{-}_{1-1-1}\vert 0\}$
11	$\{3^{-}_{-1-11}\vert 0\}$	12	$\{3^{-}_{-11-1}\vert 0\}$	13	$\{\mathrm{m}_{1-10}\vert 0\}$	14	$\{\mathrm{m}_{110}\vert 0\}$	15	$\{-4^{+}_{001}\vert 0\}$
16	$\{-4^{-}_{001}\vert 0\}$	17	$\{\mathrm{m}_{01-1}\vert 0\}$	18	$\{-4^{+}_{100}\vert 0\}$	19	$\{-4^{-}_{100}\vert 0\}$	20	$\{\mathrm{m}_{011}\vert 0\}$
21	$\{\mathrm{m}_{-101}\vert 0\}$	22	$\{-4^{-}_{010}\vert 0\}$	23	$\{\mathrm{m}_{101}\vert 0\}$	24	$\{-4^{+}_{010}\vert 0\}$		

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_1)$	A_1	0	$Q,\,T$	-	-	1
2	$\mathbb{Q}_3(A_1)$	A_1	3	$Q,\,T$	-	-	$\sqrt{15}xyz$
3	$\mathbb{G}_0(A_2)$	A_2	0	$G,\,M$	-	-	1
4	$\mathbb{G}_{2,1}(E)$	E	2	$G,\,M$	-	1	$-\frac{\sqrt{3}(x-y)(x+y)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
5	$\mathbb{G}_{2,2}(E)$					2	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
6	$\mathbb{Q}_{2,1}(E)$	E	2	Q, T	-	1	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
7	$\mathbb{Q}_{2,2}(E)$					2	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
8	$\mathbb{G}_{1,1}(T_1)$	T_1	1	G, M	-	1	x
9	$\mathbb{G}_{1,2}(T_1)$					2	y
10	$\mathbb{G}_{1,3}(T_1)$					3	z
11	$\mathbb{G}_{2,1}(T_1)$	T_1	2	G, M	-	1	$\sqrt{3}yz$
12	$\mathbb{G}_{2,2}(T_1)$					2	$\sqrt{3}xz$
13	$\mathbb{G}_{2,3}(T_1)$					3	$\sqrt{3}xy$
14	$\mathbb{Q}_{3,1}(T_1)$	T_1	3	Q, T	-	1	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
15	$\mathbb{Q}_{3,2}(T_1)$					2	$-\frac{\sqrt{15}y(x-z)(x+z)}{2}$
16	$\mathbb{Q}_{3,3}(T_1)$					3	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
17	$\mathbb{Q}_{1,1}(T_2)$	T_2	1	Q, T	-	1	x
18	$\mathbb{Q}_{1,2}(T_2)$					2	y
19	$\mathbb{Q}_{1,3}(T_2)$					3	z
20	$\mathbb{Q}_{2,1}(T_2)$	T_2	2	Q, T	-	1	$\sqrt{3}yz$
21	$\mathbb{Q}_{2,2}(T_2)$					2	$\sqrt{3}xz$
22	$\mathbb{Q}_{2,3}(T_2)$					3	$\sqrt{3}xy$
23	$\mathbb{Q}_{3,1}(T_2)$	T_2	3	Q, T	-	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
24	$\mathbb{Q}_{3,2}(T_2)$					2	$-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
25	$\mathbb{Q}_{3,3}(T_2)$					3	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$

Basis in full matrix

Table 3: dimension = 6

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\lvert p_x\rangle$ @As(1)	1	$\lvert p_y\rangle$ @As(1)	2	$\lvert p_z\rangle$ @As(1)	3	$\lvert p_x\rangle$ @Ga(1)	4	$\lvert p_y\rangle$ @Ga(1)
5	$\lvert p_z\rangle$ @Ga(1)								

Table 4: Atomic basis (orbital part only)

orbital	definition
$\lvert p_x\rangle$	x
$\lvert p_y\rangle$	y
$\lvert p_z\rangle$	z

SAMB: 48 (all 48)

- `As` : '`As`' site-cluster
  * bra: $\langle p_x\rvert$, $\langle p_y\rvert$, $\langle p_z\rvert$
  * ket: $\lvert p_x\rangle$, $\lvert p_y\rangle$, $\lvert p_z\rangle$
  * wyckoff: `4c`

$\boxed{\text{z1}}$ $\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(s)}(A_1)$

$\boxed{\text{z6}}$ $\mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$

$\boxed{\text{z7}}$ $$\mathbb{Q}_{2,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z28}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$$

$\boxed{\text{z29}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$$

$\boxed{\text{z30}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$$

- `Ga` : '`Ga`' site-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `4a`

$\boxed{\text{z2}}$ $$\mathbb{Q}_{0}^{(c)}(A_1) = \mathbb{Q}_{0}^{(a)}(A_1)\mathbb{Q}_{0}^{(s)}(A_1)$$

$\boxed{\text{z8}}$ $$\mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z9}}$ $$\mathbb{Q}_{2,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{0}^{(s)}(A_1)}{2}$$

$\boxed{\text{z31}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$$

$\boxed{\text{z32}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$$

$\boxed{\text{z33}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(s)}(A_1)}{3}$$

- `Ga;As_001_1` : '`As`'-'`Ga`' bond-cluster
  * bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
  * ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$
  * wyckoff: `16a@16e`

$\boxed{\text{z3}}$ $\mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$

$\boxed{\text{z4}}$ $\mathbb{Q}_3^{(c)}(A_1) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3}$

$\boxed{\text{z5}}$ $\mathbb{G}_0^{(c)}(A_2) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2)}{3}$

$\boxed{\text{z10}}$ $\mathbb{Q}_{2,1}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z11}}$ $\mathbb{Q}_{2,2}^{(c)}(E) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_0^{(b)}(A_1)}{2}$

$\boxed{\text{z12}}$ $\mathbb{G}_{2,1}^{(c)}(E,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3}$

$\boxed{\text{z13}}$ $\mathbb{G}_{2,2}^{(c)}(E,a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{2}$

$\boxed{\text{z14}}$ $\mathbb{G}_{2,1}^{(c)}(E,b) = -\frac{\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)}{2} + \frac{\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)}{2}$

$\boxed{\text{z15}}$ $\mathbb{G}_{2,2}^{(c)}(E,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2)}{3}$

$\boxed{\text{z16}}$ $\mathbb{Q}_{3,1}^{(c)}(T_1) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2)}{6} - \frac{\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2)}{6} - \frac{\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3} + \frac{\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{3}$

$\boxed{\text{z17}}$ $\mathbb{Q}_{3,2}^{(c)}(T_1) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2)}{6} + \frac{\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3} - \frac{\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2)}{6} - \frac{\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{3}$

$\boxed{\text{z18}}$ $\mathbb{Q}_{3,3}^{(c)}(T_1) = -\frac{\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{3} + \frac{\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{1,3}^{(b)}(T_2)}{3} + \frac{\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{3}$

$\boxed{\text{z19}}$ $\mathbb{G}_{1,1}^{(c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_0^{(b)}(A_1)}{3}$

$\boxed{\text{z20}}$ $\mathbb{G}_{1,2}^{(c)}(T_1) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_0^{(b)}(A_1)}{3}$

$$\boxed{\text{z21}}\quad \mathbb{G}^{(c)}_{1,3}(T_1) = \frac{\sqrt{3}\mathbb{M}^{(a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{0}(A_1)}{3}$$

$$\boxed{\text{z22}}\quad \mathbb{G}^{(c)}_{2,1}(T_1,a) = \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,1}(E)\mathbb{Q}^{(b)}_{1,1}(T_2)}{6} + \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{1,1}(T_2)}{6} - \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{1,3}(T_2)}{6} + \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{1,2}(T_2)}{6}$$

$$\boxed{\text{z23}}\quad \mathbb{G}^{(c)}_{2,2}(T_1,a) = -\frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,1}(E)\mathbb{Q}^{(b)}_{1,2}(T_2)}{6} + \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{1,3}(T_2)}{6} + \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{1,2}(T_2)}{6} - \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{1,1}(T_2)}{6}$$

$$\boxed{\text{z24}}\quad \mathbb{G}^{(c)}_{2,3}(T_1,a) = -\frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{1,2}(T_2)}{6} - \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{1,3}(T_2)}{3} + \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{1,1}(T_2)}{6}$$

$$\boxed{\text{z25}}\quad \mathbb{G}^{(c)}_{2,1}(T_1,b) = \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,3}(T_2)}{6} + \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,2}(T_2)}{6}$$

$$\boxed{\text{z26}}\quad \mathbb{G}^{(c)}_{2,2}(T_1,b) = \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,3}(T_2)}{6} + \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,1}(T_2)}{6}$$

$$\boxed{\text{z27}}\quad \mathbb{G}^{(c)}_{2,3}(T_1,b) = \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,1}(T_1)\mathbb{T}^{(b)}_{1,2}(T_2)}{6} + \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,1}(T_2)}{6}$$

$$\boxed{\text{z34}}\quad \mathbb{Q}^{(c)}_{1,1}(T_2,a) = \frac{\sqrt{3}\mathbb{Q}^{(a)}_{0}(A_1)\mathbb{Q}^{(b)}_{1,1}(T_2)}{3}$$

$$\boxed{\text{z35}}\quad \mathbb{Q}^{(c)}_{1,2}(T_2,a) = \frac{\sqrt{3}\mathbb{Q}^{(a)}_{0}(A_1)\mathbb{Q}^{(b)}_{1,2}(T_2)}{3}$$

$$\boxed{\text{z36}}\quad \mathbb{Q}^{(c)}_{1,3}(T_2,a) = \frac{\sqrt{3}\mathbb{Q}^{(a)}_{0}(A_1)\mathbb{Q}^{(b)}_{1,3}(T_2)}{3}$$

$$\boxed{\text{z37}}\quad \mathbb{Q}^{(c)}_{1,1}(T_2,b) = -\frac{\sqrt{30}\mathbb{Q}^{(a)}_{2,1}(E)\mathbb{Q}^{(b)}_{1,1}(T_2)}{30} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{1,1}(T_2)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{1,3}(T_2)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{1,2}(T_2)}{10}$$

$$\boxed{\text{z38}}\quad \mathbb{Q}^{(c)}_{1,2}(T_2,b) = -\frac{\sqrt{30}\mathbb{Q}^{(a)}_{2,1}(E)\mathbb{Q}^{(b)}_{1,2}(T_2)}{30} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{1,3}(T_2)}{10} - \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,2}(E)\mathbb{Q}^{(b)}_{1,2}(T_2)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,3}(T_2)\mathbb{Q}^{(b)}_{1,1}(T_2)}{10}$$

$$\boxed{\text{z39}}\quad \mathbb{Q}^{(c)}_{1,3}(T_2,b) = \frac{\sqrt{30}\mathbb{Q}^{(a)}_{2,1}(E)\mathbb{Q}^{(b)}_{1,3}(T_2)}{15} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,1}(T_2)\mathbb{Q}^{(b)}_{1,2}(T_2)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,2}(T_2)\mathbb{Q}^{(b)}_{1,1}(T_2)}{10}$$

$$\boxed{\text{z40}}\quad \mathbb{Q}^{(c)}_{1,1}(T_2,c) = \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,2}(T_1)\mathbb{T}^{(b)}_{1,3}(T_2)}{6} - \frac{\sqrt{6}\mathbb{M}^{(a)}_{1,3}(T_1)\mathbb{T}^{(b)}_{1,2}(T_2)}{6}$$

$\boxed{\text{z41}}$ $\mathbb{Q}_{1,2}^{(c)}(T_2,c) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,3}^{(b)}(T_2)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)}{6}$

$\boxed{\text{z42}}$ $\mathbb{Q}_{1,3}^{(c)}(T_2,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_1)\mathbb{T}_{1,2}^{(b)}(T_2)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_1)\mathbb{T}_{1,1}^{(b)}(T_2)}{6}$

$\boxed{\text{z43}}$ $\mathbb{Q}_{2,1}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z44}}$ $\mathbb{Q}_{2,2}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z45}}$ $\mathbb{Q}_{2,3}^{(c)}(T_2) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{0}^{(b)}(A_1)}{3}$

$\boxed{\text{z46}}$ $\mathbb{Q}_{3,1}^{(c)}(T_2) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2)}{10} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{1,1}^{(b)}(T_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{15}$

$\boxed{\text{z47}}$ $\mathbb{Q}_{3,2}^{(c)}(T_2) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{1,3}^{(b)}(T_2)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E)\mathbb{Q}_{1,2}^{(b)}(T_2)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{15}$

$\boxed{\text{z48}}$ $\mathbb{Q}_{3,3}^{(c)}(T_2) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E)\mathbb{Q}_{1,3}^{(b)}(T_2)}{5} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(T_2)\mathbb{Q}_{1,2}^{(b)}(T_2)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(T_2)\mathbb{Q}_{1,1}^{(b)}(T_2)}{15}$

Atomic SAMB

- bra: $\langle p_x|$, $\langle p_y|$, $\langle p_z|$
- ket: $|p_x\rangle$, $|p_y\rangle$, $|p_z\rangle$

$\boxed{\text{x1}}$ $\mathbb{Q}_{0}^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

$\boxed{\text{x2}}$ $\mathbb{Q}_{2,1}^{(a)}(E) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$

$\boxed{\text{x3}}$ $$\mathbb{Q}_{2,2}^{(a)}(E) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x4}}$ $$\mathbb{Q}_{2,1}^{(a)}(T_2) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$\boxed{\text{x5}}$ $$\mathbb{Q}_{2,2}^{(a)}(T_2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x6}}$ $$\mathbb{Q}_{2,3}^{(a)}(T_2) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x7}}$ $$\mathbb{M}_{1,1}^{(a)}(T_1) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$\boxed{\text{x8}}$ $$\mathbb{M}_{1,2}^{(a)}(T_1) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$$

$\boxed{\text{x9}}$ $$\mathbb{M}_{1,3}^{(a)}(T_1) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Cluster SAMB

- Site cluster

** Wyckoff: `4c`

$\boxed{\text{y1}}$ $\mathbb{Q}_{0}^{(s)}(A_1) = [1]$

** Wyckoff: `4a`

$\boxed{\text{y2}}$ $\mathbb{Q}_{0}^{(s)}(A_1) = [1]$

- Bond cluster

** Wyckoff: `16a@16e`

$\boxed{\text{y3}}$ $\mathbb{Q}_{0}^{(b)}(A_1) = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$

$\boxed{\text{y4}}$ $\mathbb{T}_{0}^{(b)}(A_1) = \left[\frac{i}{2}, \frac{i}{2}, \frac{i}{2}, \frac{i}{2}\right]$

$\boxed{\text{y5}}$ $\mathbb{Q}_{1,1}^{(b)}(T_2) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

$\boxed{\text{y6}}$ $\mathbb{Q}_{1,2}^{(b)}(T_2) = \left[\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right]$

$\boxed{\text{y7}}$ $\mathbb{Q}_{1,3}^{(b)}(T_2) = \left[\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right]$

$\boxed{\text{y8}}$ $\mathbb{T}_{1,1}^{(b)}(T_2) = \left[\frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}\right]$

$\boxed{\text{y9}}$ $\mathbb{T}_{1,2}^{(b)}(T_2) = \left[\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}\right]$

$\boxed{\text{y10}}$ $\mathbb{T}_{1,3}^{(b)}(T_2) = \left[\frac{i}{2}, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}\right]$

— Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	`As`	$\vert p_x\rangle$, $\vert p_y\rangle$, $\vert p_z\rangle$
2	`Ga`	$\vert p_x\rangle$, $\vert p_y\rangle$, $\vert p_z\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	`As`	`Ga`	`[1]`	`[p]`	`[p]`

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: '`As`' (#1) site cluster (`4c`), `-43m`

SL	position ($\boldsymbol{s}$)	mapping
1	`[ 0.25000, 0.25000, 0.25000]`	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 8: 'Ga' (#2) site cluster (4a), -43m

SL	position ($\boldsymbol{s}$)	mapping
1	[0.00000, 0.00000, 0.00000]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 9: 1-th 'As'-'Ga' [1] (#1) bond cluster (16a@16e), D, $|\boldsymbol{v}|$= 0.43301 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.25000, 0.25000, 0.25000]	[0.12500, 0.12500, 0.12500]	[1,5,9,13,17,21]	(1,1)	(1,1)	[0,0,0]
2	[-0.25000,-0.25000, 0.25000]	[0.87500, 0.87500, 0.12500]	[2,7,12,14,19,24]	(1,4)	(1,1)	[0,0,1]
3	[-0.25000, 0.25000,-0.25000]	[0.87500, 0.12500, 0.87500]	[3,8,10,16,18,23]	(1,3)	(1,1)	[0,1,0]
4	[0.25000,-0.25000,-0.25000]	[0.12500, 0.87500, 0.87500]	[4,6,11,15,20,22]	(1,2)	(1,1)	[1,0,0]